

Electricity & Magnetism

Jaysen Tsao

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5 Magnetic Fields and Electromagnetism

5.1 Introduction to Magnetism

- Magnetism is a fundamental force of nature that arises from the motion of electric charges.
- Magnetic fields are produced by moving charges (currents) and exert forces on other moving charges.
- There are two types of magnetic materials:
 - **ferromagnetic** materials (e.g. iron) can be permanently magnetized
 - **paramagnetic** (or *diamagnetic*) materials are only weakly attracted or repelled by magnetic fields.
- A **magnet** is an object that produces a magnetic field. All magnets (that we know of) have two poles: a **north pole** and a **south pole**.
 - Like poles repel each other, while opposite poles attract each other.
 - The magnetic field lines of a magnet form **closed loops** from the north pole to the south pole.
- The Earth itself is a magnet. Technically, the Earth's North pole is a magnetic south pole, for the north end of a compass needle is attracted to it.

5.2 Magnetic Fields and Magnetic Force

- Magnetic fields are produced by moving charges (currents). They exert forces on moving charges.
- The magnetic field is represented by the vector $\vec{B}$, measured in *Teslas* (T).
 - A 1 Tesla magnetic field exerts a force of 1 Newton on a 1 Coulomb point charge moving at 1 m/s **perpendicular** to the field.
 $T = N s C^{-1} m^{-1} = kg s^{-2} A^{-1}$ in SI base units.
- The direction of the magnetic field is the direction in which the north pole of a compass needle would point if placed in the field.
- **Magnetic field lines** are a visual representation of the magnetic field.
 - The direction of the magnetic field at any point is tangent to the magnetic field line at that point
 - Stronger magnetic fields have more closely spaced field lines.
 - *Unlike* electric field lines, magnetic field lines form *closed loops*!
 - *Outside* of a bar magnet, magnetic field lines point from the north pole to the south pole.
 - *Inside* of a bar magnet, magnetic field lines point from the south pole to the north pole.

Magnetic Force

Let q be a charge moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$.

The **magnetic force** $\vec{F}_b$ on the charge is:

$$\vec{F}_b = q\vec{v} \times \vec{B}.$$

- The magnetic force is always perpendicular to both the velocity of the charge and the magnetic field.
- The magnetic force does no work on the charge, since it is always perpendicular to the velocity of the charge.
- We can also calculate the magnetic force using current density $\vec{J}$:

$$d\vec{F}_b = (\vec{J} \times \vec{B}) dV \rightarrow \vec{F}_b = \iiint_E (\vec{J} \times \vec{B}) dV.$$

Lorentz Force Law

The **Lorentz force** $\vec{F}$ on a charge q moving with velocity $\vec{v}$ in both electric and magnetic fields $\vec{E}$ and $\vec{B}$ is:

$$\vec{F} = \vec{F}_e + \vec{F}_b = q\vec{E} + q\vec{v} \times \vec{B} = q(\vec{E} + \vec{v} \times \vec{B}).$$

The Lorentz force describes how charged particles behave in electromagnetic fields.

5.3 Magnetic Fields of Current-Carrying Wires

- If charge is moving in a straight line (e.g. a straight wire), then the velocity is constant and we can write the magnetic force as:

$$\vec{F}_b = \frac{q\vec{L}}{t} \times \vec{B} = I\vec{L} \times \vec{B}.$$

- ▶ Here, $\vec{L}$ is the length of a wire segment, and if it takes the charge q a time t to pass through the wire segment, then the current $I = q/t$.
- ▶ The magnetic force on a charge moving through a wire segment is proportional to the length of that segment, the current through that segment, and the magnetic field strength.
- ▶ For a wire that isn't straight, we can divide the wire into tiny segments of length $d\vec{s}$ to get:

$$d\vec{F}_b = I d\vec{s} \times \vec{B} \rightarrow \vec{F}_b = \int_C I d\vec{s} \times \vec{B}.$$

- A **coil** has no net magnetic force on it, since the magnetic forces on opposite sides of the coil cancel each other out ($\oint I d\vec{s} \times \vec{B} = 0$). However, a coil can still *produce* a magnetic field.
- A **solenoid** is a long coil of wire (a wire formed from multiple coils). The magnetic field inside a solenoid is approximately uniform and parallel to the axis of the solenoid, while the magnetic field outside a solenoid is weak and non-uniform.

Biot-Savart Law

Let I be the current through a small wire segment $d\vec{s}$. The magnetic field $d\vec{B}$ at a point P due to this wire segment is:

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{s} \times \hat{r}}{r^2}$$

where:

- μ_0 is the **permeability of free space**
- $\hat{r}$ is the unit vector from the wire segment to point P
- r is the distance from the wire segment to point P

If a wire is curved in a path C , then the total magnetic field at point P is:

$$\vec{B} = \frac{\mu_0}{4\pi} \int_C \frac{I d\vec{s} \times \hat{r}}{r^2}.$$

By the definition of the cross product, the magnitude of the magnetic field contribution from the wire segment is:

$$|d\vec{B}| = dB = \frac{\mu_0}{4\pi} \frac{I d\vec{s} \sin \theta}{r^2}$$

where θ is the angle between $d\vec{s}$ and $\hat{r}$.

Example: Magnetic Field from a Straight Current-Carrying Wire

A long, straight wire carries a current I . What is the magnetic field $\vec{B}$ at a point P a distance r away from the wire?

A long, straight wire carries a current I . What is the magnetic field $\vec{B}$ at a point P a distance r away from the wire?

Consider a small wire segment $d\ell$ at an angle θ from the horizontal axis. The distance from this wire segment to point P is $\frac{r}{\cos \theta}$. Thus, by the Biot-Savart Law:

$$dB = \frac{\mu_0}{4\pi} \frac{I d\ell \sin \theta}{\left(\frac{r}{\cos \theta}\right)^2} = \frac{\mu_0 I}{4\pi r^2} d\ell \sin \theta \cos^2 \theta.$$

5.4 Magnetic Dipoles

- *Recall*: an electric dipole consists of two equal and opposite charges separated by a small distance.
- Unlike charges which can exist independently, magnetic poles always come in pairs¹ (north and south). Thus, all magnets are **magnetic dipoles**.
- The magnetic field of a magnetic dipole is similar to the electric field of an electric dipole, but with some differences:
 - The magnetic field lines of a magnetic dipole form closed loops, while the electric field lines of an electric dipole start at the positive charge and end at the negative charge.
 - The magnetic field of a magnetic dipole decreases with distance by the cube ($B \propto 1/r^3$), while the electric field of an electric dipole decreases with distance by the square ($B \propto 1/r^2$).
- The **magnetic dipole moment** $\vec{\mu}$ of a magnetic dipole is a vector that points from the south pole to the north pole.

In a region enclosed by a current-carrying coil with area A , the magnetic dipole moment is:

$$\vec{\mu} = IA\hat{n} = I \underbrace{\vec{A}}_{A\hat{n}}.$$

where:

- I is the current through the coil
- A is the area of the coil
- $\hat{n}$ is the unit vector perpendicular to the plane of the coil, pointing in the direction given by the right-hand rule (curl your hands in the direction of current, and your thumb will point in $\hat{n}$)

For a solenoid with N loops, the magnetic dipole moment is:

$$\vec{\mu} = NI\vec{A}.$$

- A magnetic field exerts a torque $\vec{\tau}$ on a magnetic dipole:

$$\vec{\tau} = \vec{\mu} \times \vec{B}.$$

The magnetic dipole will experience a torque that tends to align it with the magnetic field.

This means that there is the potential for work to be done to align a magnetic dipole with a magnetic field, given by the potential energy U_μ of the dipole:

$$U_\mu = -\vec{\mu} \cdot \vec{B}.$$

¹ Although we aren't able to *disprove* magnetic monopoles, none have been discovered.

Magnetic Flux and Gauss's Law for Magnetism

Magnetic Flux

The **magnetic flux** Φ_B through a surface S is defined as:

$$\Phi_B = \iint_S \vec{B} \cdot d\vec{A}.$$

- The magnetic flux through a surface is a measure of the amount of magnetic field passing through that surface.
- The units of magnetic flux are Webers (Wb), where $1 \text{ Wb} = 1 \text{ T m}^2$.
- The magnetic flux through a closed surface is always zero, since magnetic field lines form closed loops. This is known as **Gauss's Law for Magnetism**:

$$\oiint_S \vec{B} \cdot d\vec{A} = 0 \quad \rightarrow \quad \vec{\nabla} \cdot \vec{B} = 0 \text{ by the Divergence Theorem.}$$

5.5 Ampere's Law

Ampere's Law

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enclosed}}$$